

Roll No.

TCS-571

B. TECH. (CSE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

BIG DATA VISUALIZATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What is Data Visualization ? Why is data visualization important ?
Explain its role with appropriate case study. (CO1)

OR

- (b) What is Big Data Visualization ? What are the various challenges in Big Data Visualization ? Explain its role with appropriate case study. (CO1)
2. (a) Condé Nast, a global media company, wanted better insight into consumer behaviour to improve the performance of its 20 industry-leading print and digital media brands. For better efficiency and business intelligence (BI), the company implemented a solution based on Visualization Tool. Design the methodology for data visualization of Condé Nast. (CO2)

P. T. O.

OR

(b) Consumer goods manufacturer Majans is using its delicious and innovative products to create a new movement in snacking. The company views technology as a key ingredient to making that happen while simultaneously unlocking employee potential and scaling for growth. Majans is transforming data from IoT-enabled devices into actionable insights on the production line to create data-driven manufacturing processes that extend from the farm to the factory to the consumer experience. Design the methodology to prepare and familiarize of data. (CO2)

3. (a) Nestlé is the largest food company in the world, and a key to its success has been its ability to balance continuity with change. The company recognized that its employees needed better access to data to react quickly to the market. The company had developed a centralized data warehouse, but it needed to combine it with local datasets to improve decision-making and provide sustainable organic growth. Analyse the importance of Editorial Focus in the perspective of Nestlé. (CO2)

OR

(b) Lenovo, a global technology company, aimed to optimize its analytics experience across all the departments and worldwide offices. Previously, Lenovo operated with one single sales report that was delivered to 28 different countries. When different regions or company's divisions wanted to adopt the report to extract the most valuable data, it required a commitment of eight to ten individuals and

led to a massive number of on-hold tasks for the analytics team. In turn, Lenovo decided to use Tableau to orderly structure the data all across the company. As a result, Lenovo got a flexible dashboard with all the sales that can be adapted for the ad-hoc analyses, which also led to 95% efficiency improvement across 28 countries. With the help of Tableau dashboard ideas, Lenovo gathered the engagement metric, thus crafting a better experience and collecting more revenue. Refine the editorial focus of visualization tool in the perspective of Lenovo. (CO2)

4. (a) J. P. Morgan Chase & Co. is a leading multinational bank and financial services company based in the USA. It is the largest investment bank in the US and sixth largest in the world. The amount of data produced by the company increased with increasing growth and expansion of business of the firm because of successful mergers and acquisitions. This led to the need for a robust self-service data governance and analytics solution. To boost the efficiency and carry out a real-time data, J. P. Morgan Chase & Co. adopted Tableau. Compare and contrast Traditional BI Tools and Tableau. (CO2)

OR

- (b) Lufthansa is a global aviation company which has used Tableau for its growth and benefit. Around 2016, Lufthansa group felt the need for a more centralized, standardized and uniform reporting solution. Because what they had at that time was not a uniform group reporting system. State some ways to improve the performance of Tableau in the perspective of Lufthansa. (CO2)

P. T. O.

5. (a) Fantasy Football is a rapidly growing online social game. As users become more sophisticated and technology advances, the amount of data that is available to inform users' decision-making is growing rapidly. Representing this data in an informative and engaging way can be a challenge but data visualisation offers many ways to achieve this. Identify the Charts/Graphs/Maps those are useful to track the virus in real-time, predict patterns, devise effective strategies for Fantasy Football. (CO1)

OR

- (b) Cricket is one of the most popular sports in the world with an estimated fan following of 2.5 billion people. There are primarily three types of international cricket matches-test matches, one day internationals and Twenty-20. Among these, one day internationals (ODIs) are the most commonly played type of matches in international tournaments including the ICC Cricket World Cup and the ICC Champions Trophy. Choose the optimal Graph/Chart to visualize the score of each player and justify the reason to choose. (CO1)